

Requirement Specification Document

Restaurant Management Database System

Week 1 – Requirement Analysis and Customer Database Modul

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Project Title	Restaurant Management Database System
Week	Week 1

1. Introduction

1.1 Purpose

The **Restaurant Management Database System (RMDS)** is designed to manage restaurant operations such as customer information, table reservations, food orders, billing, and menu management. The system provides a centralized database that improves operational efficiency, reduces manual work, and enhances customer service.

1.2 Project Scope

The system includes customer registration, table reservation management, order processing, billing, menu management, and report generation. It is intended for small and medium-sized restaurants.

1.3 Intended Users

-  **Customers** – Make reservations and place food orders.
-  **Restaurant Staff** – Manage orders, tables, and billing.
-  **Managers/Administrators** – Manage customer records, menu items, staff accounts, and business reports.

2. Business Requirements

The following are the key business needs the system must satisfy:

- Maintain accurate and up-to-date customer records
- Maintain a catalog of products with price and stock quantity
- Allow customers to place orders for one or more products
- Track the status of each order (Pending, Shipped, Delivered, Cancelled)
- Record payment details for each order
- Generate reports such as total sales, top-selling products, and pending orders

3. Functional Requirements

Functional requirements describe what the system must do.

ID	Requirement Description
FR-1	The system shall allow customers to register with contact details.
FR-2	The system shall allow customers to reserve tables online or at the restaurant.
FR-3	The system shall record food orders placed by customers.
FR-4	The system shall calculate bills automatically including taxes and discounts.

FR-5	The system shall maintain menu item details such as name, price, and category.
FR-6	The system shall generate daily and monthly sales reports.
FR-7	The system shall allow administrators to add, update, and remove menu items.
FR-8	The system shall provide secure login access for staff and administrators.

4. Non-Functional Requirements

- **Performance:** The system should process orders within 2 seconds.
- **Security:** Customer and billing data must be protected from unauthorized access.
- **Reliability:** The system should maintain accurate order and billing records.
- **Usability:** The interface should be easy for staff and customers to use.
- **Scalability:** The database should support increasing numbers of customers and orders.

5. Customer Database Module – Data Requirements

Since this week's focus includes the Customer Database Module, the following entity and attributes have been identified:

Attribute	Data Type	Description
customer_id	INT (Primary Key)	Unique customer identifier
full_name	VARCHAR(100)	Customer's full name
email	VARCHAR(100)	Customer email address
phone_number	VARCHAR(15)	Customer contact number
address	VARCHAR(255)	Customer address
registration_date	DATE	Date of table reservation

6. Assumptions and Constraints

- The system will use a relational database such as MySQL or PostgreSQL.
- Only authorized staff and administrators can modify restaurant data.
- Customers must provide valid contact details for reservations.
- Internet access is required for online reservations.
- This document covers **Week 1 – Requirement Analysis and Customer Database Module** only.

7. Conclusion

This Requirement Specification Document defines the business, functional, and database requirements for the **Restaurant Management Database System**. The Customer Database Module provides the foundation for designing the database schema, ER diagram, SQL tables, and future implementation of the complete restaurant management system.